



The University of Virginia

College of Arts & Sciences / Department of Statistics

STAT 3080 - From Data to Knowledge

Section (ID): 001, 002, 003 (11365, 12763, 19923)

Spring 2026

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Classroom:	Monroe Hall 110 (001, 002) Nau Hall 211 (003)

Course Description

Catalog Description

STAT 3080 - From Data to Knowledge

Effective Start Date: 08/01/2024

This course introduces methods to approach uncertainty and variation inherent in elementary statistical techniques from multiple angles. Simulation techniques such as the bootstrap will also be used. Conceptual discussion in lectures is supplemented with hands-on practice in applied data-analysis tasks using R. Prerequisite: A prior course in statistics and a prior course in programming.

Requisites Students must have completed ONE of STAT 1100, STAT 1120, STAT 2020, STAT 2120, STAT 3120, APMA 3110, APMA 3120 & ONE of STAT 1601, STAT 1602, STAT 3250, CS 1110, CS 1111, CS 1112, CS 1113

Credits: 3

Course Attributes: Quantification, Computation & Data Analysis

Instructional Mode

This is an [In Person](#) course.

Course Goals and Learning Objectives

The student should be able to:

1. Follow [literate programming](#) principles to write and debug basic [R](#) scripts that are easily read by a human and executable on a personal computer.
2. Effectively communicate in writing using [reproducible research](#) principles and tools including [knitr](#).
3. Use simulation to study operating characteristics of parametric and non-parametric one and two-sample inferential procedures under various data scenarios.
4. Use simulation to make practical decisions when assumptions of parametric and non-parametric procedures cannot be verified.
5. Understand and employ elementary version control principles and tools including [Git](#).

This course focuses on the simulation-based inference approach to concepts taught in elementary statistics (roughly equivalent to AP Statistics). This course will not teach you how to:

1. Work with sound, image, or other non-text data.
2. Deploy artificial intelligence (AI) pipelines.
3. ETL using industry-scale parallel/cluster/cloud systems.

4. Achieve world peace or conquer the stock market using neural networks.

There are other courses, mostly outside of the Statistics Department, that achieve these objectives. You are encouraged to take them.

Course Materials

There will be assigned readings from nine different sources as well as recorded pre-lecture videos posted on Canvas. Ideally you will read the text and watch the videos *before* attending class. All materials are either freely available online or will be posted on Canvas.

Required

Garrett Golemund. *Hands-On Programming with R: Write Your Own Functions and Simulations*. O'Reilly Media, Sebastopol, CA, 2014. ISBN 978-1449359010. <https://rstudio-education.github.io/hopr/> Accessed: 2026-01-03

Taylor R. Brown. *An Introduction to R and Python For Data Analysis: A Side By Side Approach*. CRC Press Taylor & Francis Group, Boca Raton, FL, 1st edition, 2023. ISBN 9781032203256. <https://tbrown122387.github.io/publication/randpybook.html> Accessed: 2025-08-22

Hadley Wickham, Mine Cetinkaya-Rundel, and Garrett Golemund. *R for Data Science*. O'Reilly, Sebastopol, CA, 2nd edition, 2023. ISBN 978-1492097402. <https://r4ds.hadley.nz/> Accessed: 2025-08-26

Hadley Wickham and Garrett Golemund. *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. O'Reilly Media, Inc., Sebastopol, CA, USA, 2017. ISBN 978-1491910399. doi: 10.5555/3086927. <https://r4ds.had.co.nz/> Accessed: 2026-01-03

Roger D. Peng. *R Programming for Data Science*. Leanpub and lulu.com, 5th edition (ebook online edition) edition, 2022. ISBN 978-1365056826. <https://bookdown.org/rdpeng/rprogdatascience/> Accessed: 2026-01-03

Jerry Bonnell and Mitsunori Ogihara. *Exploring Data Science with R and the Tidyverse: A Concise Introduction*. Chapman & Hall/CRC Data Science Series. Chapman & Hall / CRC Press, 2023. ISBN 9781032341705. doi: 10.1201/9781003320845. <https://ds4world.cs.miami.edu/> Accessed: 2026-01-05

Clay Ford. *Getting Started With the pwr Package*, 2020. <https://cran.r-project.org/web/packages/pwr/vignettes/pwr-vignette.html> Accessed: 2026-01-03

Additional Resources

Scott Chacon and Ben Straub. *Pro Git*. APress, 2nd edition, 2014. ISBN 978-1484200773. <https://git-scm.com/book/en/v2> Accessed: 2025-08-22

Layth C. Alwan, Bruce A. Craig, George P. McCabe, and David S. Moore. *The Practice of Statistics for Business and Economics*. W.H. Freeman and Company, New York, fifth edition, 2020. ISBN 1319324819

Paul H. Kvam and Brani Vidakovic. *Nonparametric Statistics with Applications to Science and Engineering*. Wiley, 2007. ISBN 978-0470081471. doi: 10.1002/9780470168707

Software (Required)

We will utilize [R](#), a statistical software system developed under an open source license, making it freely usable and distributable. To demonstrate R code, I will often, though not exclusively use [RStudio](#), which is a free integrated development environment (IDE).

Source Control Management (SCM)

You are encouraged to manage your code using [Git](#) and [choose between](#) two industry standards: [GitHub](#) and [Bitbucket](#) to host your remote repository. You *must use GitHub to submit your project deliverables*, but you are not required to manage your daily assignments or homework source code. You are encouraged to use the command line interface (CLI); however, there are many popular [GUIs](#). One popular option is [Visual Studio Code](#) (VSCode), which is a Windows-based IDE that integrates seamlessly with [Git/GitHub](#) and [Bitbucket](#). [Bitbucket+Sourcetree](#) is another option. I will also use [Positron](#), a fork of VSCode developed by [Posit](#) (maker of [RStudio](#)). Positron is a "polyglot" IDE that supports R, Python, and SCM in addition to many other industry-level data science tools. This is the tool I use in STAT 3250.

Hardware (Required)

While this course does not require prior R programming experience, it does require that you know how to operate a personal computer. You are responsible for having access to a device upon which you can complete assignments. If you have a laptop, feel free to bring it to lecture in order to follow along with demonstrations or work on [assignments](#).

Assignments

There are 3 categories of assignments: Homework, Project, and Exam. The textbooks and recorded pre-lecture videos will be our primary sources, and assignments will allow you to sharpen your understanding of the topics covered in these sources.

Homework assignments will be hosted on the course GitHub repository. Assignments will focus on the details of manipulating and analyzing data. You will submit your solutions by uploading a single `.Rmd` file (and image files for select assignments) to [Gradescope](#), which will be linked to our course Canvas. Gradescope will autograde your code using unit tests, and TAs will grade your free responses. You will have unlimited opportunities to resubmit your file(s) prior to the deadline. You may use *any* resource to assist you in completing homework assignments, including your classmates, [Stackoverflow](#), and artificial intelligence such as [ChatGPT](#), [GitHub Copilot](#), or [Cursor AI](#). However, if you submit another student's solutions as your own, you will earn a 0

on that assignment. Homework assignments are always due at 11:59 PM on the date listed in the course schedule.

There will be three full-length Exams administered in paper form during the lecture period. A practice Exam of the same format and length (but not necessarily the same questions) as the actual Exam will be provided. You will be permitted to use 1 double-sided sheet of notes when taking an Exam, but you will not be permitted to use a textbook or electronics (in particular, calculators are prohibited). Exam questions will require you to write and interpret syntactically correct R code. As detailed in the [Course Schedule](#) on page 9, each Exam will focus on R and statistical concepts presented after the preceding Exam, but syntax builds upon itself so the Exams are not completely independent.

There is no final exam in this course. Instead, there will be a half-semester long project due in two parts: Part 1 on 04/12/26 and Part 2 on the last instructional day, 04/28/26. Projects can be completed in teams of up to three. Please refer to the course Canvas for additional details.

Grading Policy

During the semester you may earn up to 100 points through assignments (Homework, Project, and Exam). There will be 11 Homework Assignments, 1 Project, and 3 Exams. The lowest score in each category (except for the project since there is only 1 project) will be dropped before computing your total points for a category. Your cumulative total for the semester will be the sum of your total points in each category. Grades will be maintained on the course Canvas. A grading rubric for Homework assignments will be automatically provided in the Gradescope autograder.

Component	Point Value Per Item	Number of Items	Item Dates	Total (After Lowest Dropped)
Homework	4	11	See Schedule on page 9	40
Project	20	1	04/12, 04/28 (2 parts)	20
Exam	20	3	02/25, 04/06, 04/22	40
Cumulative				100

I will assign your final letter grade according to which interval your cumulative point total falls in the following table. Your point total will be rounded up to the nearest whole number before computing your cumulative total for the semester. I reserve the right to assign a letter grade that is *higher* than what is given in the table, for example, if cumulative point totals on the exam are lower than expected.

Cumulative Points	Letter Grade
[100 , 90]	A
[89 , 87]	B+
[86 , 80]	B
[79 , 77]	C+
[76 , 70]	C
[69 , 67]	D+
[66 , 60]	D
[59 , 00]	F

Late/Missed Work and Make-Up Policy

Barring an "excusable reason," a late Homework submission will be penalized 3 points per day and an Exam cannot be made up. Please refer to the [university-wide](#) and [college-specific](#) policies governing late/missed work, including what constitutes an excusable reason. Please communicate with me as soon as possible if you intend to make up late or missed assignments.

Course Policies

Classroom Protocol

Your classroom behaviour should contribute to a respectful, welcoming, and inclusive environment. Classroom discussion should be civilized and respectful to everyone, and discussion from class that continues in an online medium should adhere to these same expectations. Please ensure that your cell phone use does not disrupt your classmates' learning.

Communication Guidelines

You are responsible for knowing course-related information that I communicate via your university email account and the Canvas course page. It would be helpful to me if you begin the subject line of your emails with "STAT 3080".

Academic Integrity

Collaborating with classmates and utilizing outside sources is permitted on all assignments *except for* Exams. On the Exams you may not use any natural or artificial intelligence, other than your own. If you violate this exam policy, you will earn a 0 on the exam and cannot earn higher than a C- in the course.

Religious Accommodations

The course aims to support students observing religious practices, including fasting during periods of religious significance (i.e., Ramadan, Passover, and so on). Students may request:

- a 12-hour extension on assignments due during the period of observance.
- same-day rescheduling of exams to accommodate their needs (e.g., moving exams to earlier in the day).

To ensure arrangements can be made, please submit your requests in writing at least 7 days in advance where possible. These accommodations are intended to create a fair and supportive environment while maintaining the integrity of the course schedule.

Academic Regulations

Regulations governing Attendance, Final Examinations, Grading, and Registration & Withdrawal can be found under the [Academic Regulations](#) section of the most recent *Undergraduate Record* at <http://records.ureg.virginia.edu/index.php>. Regulations governing Academic Integrity can be found under the [Honor System](#) section of the *Undergraduate Record*.

Add/Drop Deadline

Guidelines governing adding, dropping, or withdrawing from a course can be found at https://college.as.virginia.edu/add_drop_withdraw. The Add/Drop deadline can be found at <https://college.as.virginia.edu/calendar>.

Course Materials

[University policy](#) limits re-distribution of classroom materials (e.g., lectures, syllabi, lecture notes, exams, problem sets, and presentations) in order to respect the integrity and effectiveness of the classroom experience; protect students and faculty dignity and privacy; respect faculty and University rights in instructional materials; and comply with copyright law.

Accessibility

UVA is committed to creating a learning environment that meets the needs of its diverse student body. If you anticipate or experience any barriers to learning in this course, please feel welcome to discuss your concerns with me. If you have a disability, or think you may have a disability, you may also want to meet with the [Student Disability Access Center \(SDAC\)](#) to request an official accommodation. You can find more information about SDAC, including how to apply online, through their [website](#). If you have already been approved for accommodations through SDAC, please make sure to send me your accommodation letter and meet with me so we can develop an implementation plan together.

Final Examination

Final Exam rescheduling policies and procedures can be found at <https://college.as.virginia.edu/finalexams>. Additional policies can be found at https://uvapolicy.virginia.edu/policy/prov-008#Final_Examinations.

COVID-19 Compliance, Classroom, and Campus Safety

Please abide by all campus safety protocols as outlined in the most recent campus updates at <https://coronavirus.virginia.edu/>.

Student Support Resources

The University of Virginia recognizes that students juggle different responsibilities and can face challenges that make learning difficult. There are many resources available to help students navigate academic, emotional/psychological, physical/medical, material/accessibility concerns, including:

- [ITS](#) (Technical Support) for accounts, passwords, email, networks, etc
- [Tutoring](#) for multiple subjects and writing center
- [Just Report It](#) is UVA's gender-based violence, assault, hazing, and discrimination reporting system
- [National Suicide Prevention Hotline](#) 1-(800)-273-8255
- For a list of other resources available to students, see <https://college.as.virginia.edu/university-resources>.

Course Schedule

While I will make every effort to follow the schedule below, we may slightly deviate based on the difficulty of the topics. Changes to this schedule will be announced by email and/or on Canvas. B = [Brown, 2023], G = [Grolemund, 2014], W = [Wickham and Grolemund, 2017], WC = [Wickham, Cetinkaya-Rundel, and Grolemund, 2023], P = [Peng, 2022], BO = [Bonnell and Ogihara, 2023], A = [Alwan, Craig, McCabe, and Moore, 2020], V = [Kvam and Vidakovic, 2007], F = [Ford, 2020]. You are not expected to know all the details in [V] since it is fairly technical.

Week	Date	Activity	Coverage
1	01/12	Read Syllabus; Preface, Ch 1, 2 [B]; Appendix, Ch 2 [G]	Installation, R Basics; Installation, R Basics
1	01/14	Read 27 [W]	Markdown
2	01/19	MLK Day Observed	<i>No Class</i>
2	01/21 01/25	Read Ch 3 (skip 3.9) [B]; Ch 5 [G] Syllabus Quiz Due	Vectors; Types
3	01/26	Read Ch 4 [B]	Matrices
3	01/28 02/01	Read Ch 5 [B] Homework 1 Due	Lists
4	02/02	Read Ch 7, 8 [B]	Factors, Dataframes
4	02/04 02/08	Read 6 [G] Homework 2 Due	Subset, Select
5	02/09	Read Ch 12 [B]; Ch 6, 11 [B]	Sort, Merge; Functions, Logic, Loops
5	02/11 02/15	Read Ch 9, 11 [G] Homework 3 Due	Functions, Logic, Loops

Week	Date	Activity	Coverage
6	02/16	Read Ch 4 [BO]	Simulation
6	02/18	Ch 12 [G]; Ch 16 [P]	Vectorized Code; apply Family
7	02/23	Catch-Up & Review	
7	02/25	Exam 1	
	02/28 – 03/08	Spring Recess	<i>No Class</i>
8	03/09	Read Ch 3 [WC]	dplyr
8	03/11 03/15	Read Ch 1 [WC] Homework 4 Due	ggplot
9	03/16	Read Ch 10, 11 [WC]	EDA
9	03/18 03/22	Read 5, 9 [WC] Homework 5 Due	Tidying, Layers
10	03/23	Read Ch 5, § 8.4 [BO]	Sampling, CLT
10	03/25 03/29	Read § 18.6 [A]; Ch 6 [BO] Homework 6 Due	Randomization Tests; Testing

Week	Date	Activity	Coverage
11	03/30	Read Ch 18 [A]; Ch 7 [BO]	Bootstrap; Resampling, Bootstrap
11	04/01 04/05	Catch-Up & Review HW 7 Due	
12	04/06	Exam 2	
12	04/08 04/12	Read Ch 14 [A]; Read Ch 7 [V] Homework 8 Due Project Part 1 Due	Nonparametric Tests; Nonparametric Tests
13	04/13	Read § 9.1 – 9.4 [V]	Categorical Data
13	04/15 04/19	Read All [F] Homework 9 Due	Power, Effect Size
14	04/20	Catch-up & Review	
14	04/22 04/26	Exam 3 Homework 10 Due	
15	04/27 04/28	Catch-up Homework 11 Due Project Part 2 Due	